

Smart Home Builder · Premium Course

3 live sessions · 2 hrs each · 10-14 yrs · from KES 9,000

Premium course outline **PREMIUM**

Overview

Kids build a working smart home model with three integrated systems, wired into one house.

DETAIL

Track	Arduino Robotics & Electronics
Format	3 live sessions · 2 hrs each
Delivery	Live online (1:1 or small group) or in person, you choose when booking
Ages	10-14 yrs
Level	Intermediate → Advanced
Price	from KES 9,000
Maker Kit	Optional, KES 8,500 one-time: Arduino Uno R3 board + USB cable, Breadboard + 40 jumper wires, LEDs (red, yellow, green, white, RGB) + resistors, Push buttons, potentiometer and piezo buzzer, Light sensor (LDR) and sound sensor, HC-SR04 ultrasonic sensor + bracket, IR sensor, DHT11 temperature & humidity sensor, PIR motion sensor, SG90 servo motor, Relay module, 16x2 LCD screen (I2C), Battery holder, Storage box. Bought once and reused in every Arduino course. Needed at home for live online sessions; for in-person sessions we provide kits, so buy one only to keep building at home.
Parts pack	Optional, KES 1,000: Small 5 V fan, Servo door-lock bracket, Cardboard house template
Standalone	No
Prerequisites	Traffic light simulation OR Button controlled LED (free)

Programme

1. **Smart Lighting.** Photoresistor + relay auto-lights rooms at dusk; a PIR motion sensor activates hallway lights.
2. **Climate Control.** DHT11 temperature sensor triggers a fan via relay; an LCD displays the current temperature.

3. **Security System.** Ultrasonic door sensor + buzzer alarm + servo-powered door lock, all wired into one house model.

What they take home

 Cardboard smart house model with all circuits installed

Why parents love it

“Your child builds a miniature version of the smart home tech you see advertised — and actually understands how it works.”

How to book

Places are limited and run on scheduled dates. To book or ask a question, email **tinkerwithanne@gmail.com** with the course name (**Smart Home Builder**), your child's age, and your preferred dates. Prices start from the amount above and may vary with group size and materials.